Ideal Gas and Viscosity Experiments

04 - October - 2024

25 - October - 2024

Experimental report submitted for the module

EMS410U - Experimental Design and Practice 1

Group 16

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Section 1 - Ideal Gas

Introduction

Abstract

The purpose of this experiment is to evaluate how well air behaves under controlled circumstances in relation to ideal gas behaviour, which is determined by the ideal gas equation, $PV=nRT$. We investigated the connection between pressure, volume, and temperature by altering air-bleed rates in a two-cylinder configuration to see if near-isothermal conditions could be attained. With a calculated volume ratio of 1.885, which is only 2.42% off the manufacturer's specified ratio of 1.84, the results demonstrated that air behaved similarly to an ideal gas at a slow air-bleed rate. This implies that, under controlled circumstances, air can behave similarly to an ideal gas, with only slight variations brought on by variables like temperature fluctuations and equipment constraints.

Background.

The ideal gas law, which combines Boyle's Law, Charles's Law, Gay-Lussac's Law, and Avogadro's Law, describes the relationship between pressure (P), volume (V), temperature (T), and the number of moles of a gas (n):

$$PV=nRT$$

where R is the ideal gas constant, ideal gases are theoretical constructs that assume no interactions between gas molecules and that the gas particles occupy no volume. While real gases deviate from these assumptions, they can often approximate ideal gas behaviour under certain conditions, such as low pressure and high temperature.

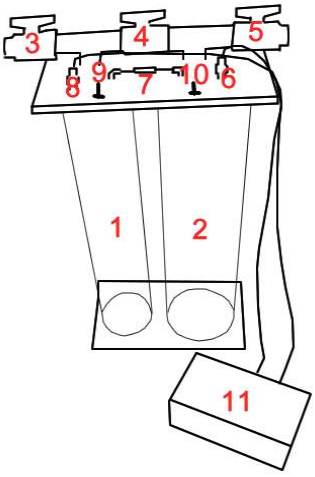
To determine if air behaves in line with the ideal gas law in near-isothermal conditions, it is treated as an ideal gas in this experiment. To replicate varied rates of expansion and compression, we used a two-cylinder arrangement with three variable air-bleed rates: rapid, medium, and slow. We aimed to determine the volume ratio between the cylinders and compare it to the manufacturer's standard by regulating the temperature and pressure throughout these rates. By using this method, we were able to investigate how well ideal gas laws may describe actual gases like air, particularly in carefully regulated environments with minimal temperature variations.

Experimental Method

Equipment

- The apparatus consists of two large cylinders connected by pipes and valves. The pipes are connected so that air can pass between the two cylinders as well as the outside air.

- The airflow in the pipes is controlled by three large valves. A needle valve between the two cylinders allows for fine control of air transfer between the cylinders. This is illustrated in the diagram below.

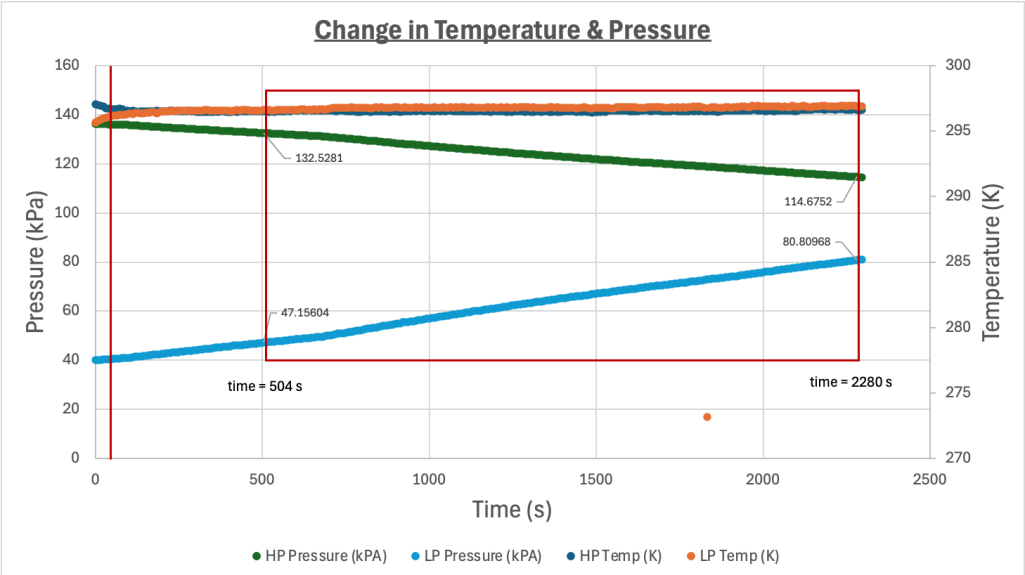


1. High-pressure cylinder //2. Low-pressure cylinder // 3. Low-pressure cylinder ball valve // 4. Ball valve // 5. High-pressure cylinder ball valve // 6. High-pressure cylinder pump coupling // 7. Needle valve // 8. High-pressure cylinder // 9. Low-pressure cylinder thermocouple // 10. High-pressure cylinder thermocouple // 11. Pressure control and display console

Results

Having carried out the experiment using 3 different air-bleed rates as explained above, we have concluded that the slow-bleed rate run has provided us with the most valid results because the observed temperatures remain almost constant throughout the experiment, meaning that the process is the closest to being isothermal.

Below is the variation of temperature and pressure in both cylinders over time. The “HP” and “LP” measurements are for the High-Pressure and low-pressure cylinders, respectively.



We have decided to use a data range of between 504s and 2280s, as shown on the above graph (using a red rectangle) to make our calculations. Hence, we will assume our ‘Initial’ state to be at 504s and our ‘Final’ state to be at 2280s.

At the initial stage, the temperatures in both Cylinders should be equal, and assuming that the experiment is truly isothermal and that the pressures in the 2

cylinders have not fully equalised, then the temperatures of the 2 cylinders should also be equal at the final state.

Given the Ideal Gas law, $PV = mRT \quad or \quad Pv = RT$

where

P = absolute pressure
T = absolute temperature
v = specific volume (mass m/volume V)
R = specific gas constant

Since we know that the Volume of the 2 cylinders remains constant and that the mass of the air in both cylinders combined also remains the same, we can rearrange the ideal gas law formula to calculate the mass of the air in each cylinder at both the initial and final states.

$$m_{\mathrm{L1}} = \frac{V_{\mathrm{L}}P_{\mathrm{L1}}}{RT_1} \qquad m_{\mathrm{H1}} = \frac{V_{\mathrm{H}}P_{\mathrm{H1}}}{RT_1} \qquad m_{\mathrm{L2}} = \frac{V_{\mathrm{L}}P_{\mathrm{L2}}}{RT_2} \qquad m_{\mathrm{H2}} = \frac{V_{\mathrm{H}}P_{\mathrm{H2}}}{RT_2}$$

Where variables with subscript “L” or “H” correspond to values for the low and high-pressure cylinders respectively, and where variables with subscript “1” or “2” refer to values for the initial and final state.

Since, $m_{\mathrm{L1}} + m_{\mathrm{H1}} = m_{\mathrm{L2}} + m_{\mathrm{H2}}$, substituting the above formulae and simplifying them with $T_1 = T_2$ (since we are assuming an isothermal process), we should be able to calculate the volume ratio between the two cylinders using:

$$V_{\mathrm{R}} = \frac{V_{\mathrm{H}}}{V_L} = \frac{P_{\mathrm{L2}} - P_{\mathrm{L1}}}{P_{\mathrm{H1}} - P_{\mathrm{H2}}}$$

Using this formula, and our results, we can calculate a volume ratio of 1.885052, which we can compare to the volume ratio supplied by the manufacturer of 1.84, to calculate a percentage difference of 2.418851%

Discussion

Analysis & Discussion

For the fast bleed rate, the minimum and maximum recorded temperatures of the cylinders, within the used data range of 104s to 192s is 292.13907K and 301.19767K respectively. This means that the temperatures remained within a significantly large 9.0586K range, which means that this particular run of the experiment can definitely not be considered as an isothermal process

For the medium bleed rate, the minimum and maximum recorded temperatures of the cylinders, within the data range of 400s to 1400s is 295.69766K and 297.57657K respectively (ignoring a single outlier value). This means that the temperatures remained within a 1.87891K range. Although this is a smaller temperature range, it is still a significant enough difference to affect our calculations.

For the slow bleed rate, the minimum and maximum recorded temperatures of the cylinders, within the previously specified data range (504s to 2280s) is 296.42032K and 296.95157K respectively (ignoring a single outlier value). This means that the temperatures remained within a small 0.53125K range. This is the closest we could get to a truly isothermal process.

The Volume ratio we have obtained from our experiment (Using data from the slow bleed rate run) is 1.885052, which we can compare to the volume ratio supplied by the manufacturer of 1.84, to calculate a percentage difference of 2.418851%.

Conclusion

The ideal gas experiment effectively assessed whether, in controlled circumstances, air acts in accordance with the ideal gas law. The experiment, which used a range of air-bleed rates, showed that the slow-bleed rate produced the most precise findings while maintaining a nearly isothermal process and a relatively constant temperature. With a percentage difference of 2.42%, the computed volume ratio between the two cylinders was 1.885, which was quite near to the manufacturer's recommended ratio of 1.84. This minor difference implies that although air does not behave exactly like an ideal gas, it does behave approximately that way under the experimental setup.

The experiment emphasises the importance of having conditions close to isothermal to see ideal gas behaviour. Although possible sources of error, such as temperature variations and instrument precision, could have impacted the findings, the results aligned with theoretical assumptions. Overall, the experiment was successful and confirmed that, in some circumstances, the ideal gas law can be used as a near approximation for air.

Future research might include prolonging the experiment by making the air bleed rate even slower. This would allow us to collect a greater amount of data and ensure that the experiment is held in a condition that is even closer to being truly isothermal.

If we had access to more resources, we could upgrade the apparatus setup to reduce mistakes and increase the accuracy of the findings. For example, we could do this by using more accurate sensors for recording temperature and pressure or by using 2 or more sensors for each measurement so that we can identify anomalies more easily and reduce the impact of random fluctuations and outliers by taking an average of the measurements.

Section 2 - Viscosity

Standard Operating Procedures(SOP)

Scope

This experiment aims to determine the viscosity of a Newtonian liquid (glycerine) by observing the motion of a smooth sphere as it falls through the liquid. The viscosity is calculated using Stoke's law: $F = 6\pi r\eta v$ where r is the radius of the sphere, η is the viscosity of the liquid, and v is the terminal velocity. The setup involves a graduated cylinder filled with glycerine, into which three metal spheres of varying diameters are dropped. The terminal velocity is determined using displacement (measured with a ruler) and time (measured with a stopwatch). These measurements are used to calculate the viscosity of glycerine.

Ball Density Standard Operating Procedure

1. **Measuring diameters of metal balls.**
   - 1.1. Gather metal spheres of three distinct diameters: **8 mm, 5.5 mm, and 3.5 mm** (three spheres for each diameter).
   - 1.2. Measure the diameter of each sphere using **both a digital and an analogue micrometre**. This dual approach ensures the identification of systematic errors, such as calibration drift in the digital micrometre or wear in the analogue micrometre.
   - 1.3. Take four measurements for each ball (two from each micrometre).
   - 1.4. Record all measurements systematically in an Excel sheet, categorizing them by diameter class (8 mm, 5.5 mm, and 3.5 mm).
   - 1.5. Calculate the **average diameter** for each sphere size to improve accuracy and reduce random errors.
2. **Measuring the mass of metal balls.**
   - 2.1. Use a **single calibrated digital balance** to measure the mass of all nine spheres. This minimizes inter-device variability.
   - 2.2. Ensure the digital balance is zeroed before each measurement to avoid offset errors.
   - 2.3. Place each ball on the balance individually and wait for the displayed reading to stabilize before recording the value.
   - 2.4. Divide each recorded weight by the acceleration due to gravity (g=9.81 m/s^2) to calculate the mass of each ball.
   - 2.5. Input all mass data into the Excel sheet for further calculations.
3. **Calculate the density of metal balls.**
   - 3.1. Use the formula for the volume of a sphere: $V = \frac{4}{3}\pi r^3$. where the **radius (r)** is half of the average diameter calculated in Step 1.5.
   - 3.2. For each sphere, calculate its volume using the measured average diameter.
   - 3.3. Determine the density of each sphere by dividing its calculated mass (Step 2.4) by its calculated volume: ρ=m/V
   - 3.4. Summarise the density results for all spheres in the Excel sheet.

Errors in measurement

We have identified these sources of both systematic and random error in the liquid density measurements, as well as steps taken to reduce these errors:

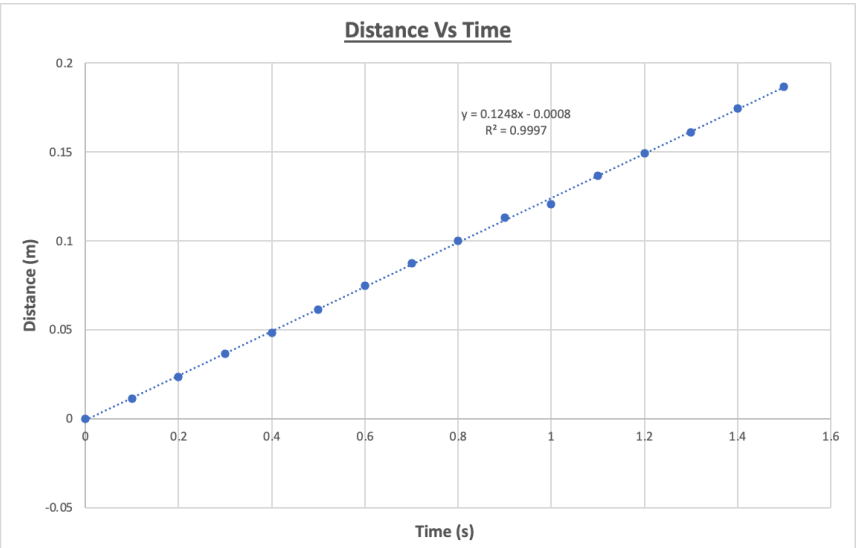
1) Due to the adhesive nature of the Glycerine solution, The liquid inside the measuring beaker is difficult to empty, and often sticks to the inner surface of the beaker, falling very slowly. This means that every time we empty and refill the beaker, a small amount of the glycerine solution will not. This can lead to systematic error. We can reduce this error by waiting longer before taking measurements to give the solution more time to fill the bottom of the beaker as much as possible.
2) When reading off the value on the beaker, it is important to read the scale at eye level to minimise reading error caused by Parallax. It is also essential to ensure that we read the value from the meniscus of the liquid.
3) We must repeat the measurements several times to identify anomalous readings caused by random error so that we can discard them.
4) Since multiple groups were using the mass scale, we recorded the mass of our empty beaker, and every time we took a measurement, we subtracted this from the value for mass. This is to avoid zero error since the beakers used by different groups have different masses.

Regression

For the viscosity calculations, we selected the **smallest metal ball (3.5 mm diameter)** as it provided the most reliable data for regression analysis, Specifically:

- The R^2 value for the regression curve of the 3.5 mm ball was **close to 1**, indicating an excellent linear fit.
- The consistency of the results from the 3.5 mm ball minimised potential errors and increased the accuracy of the derived viscosity.

To ensure reliability, we used the **average terminal velocity** from the three 3.5 mm metal balls. This approach reduced random errors and provided a robust dataset for calculating glycerine's viscosity.



By performing linear regression using the regression tool on Excel, we've determined the mean value of the Terminal velocity to be **0.12479 ± 0.00059889 m/s (5sf)**.

Now, using the below equation, we can calculate the Viscosity of the fluid using our values for Terminal Velocity, Density of the fluid, density of the ball bearing, and radius of the ball bearing.

$$\eta = \frac{2gr^2(\rho_s - \rho_l)}{9v}$$

	Obtained Value	Error
Terminal Velocity, v (m/s)	0.124794118	0.000598888
Ball Density, ϱs (kg/m³)	51526.04612	1111.258661
Fluid Density, ϱl (kg/m³)	1216.223577	11.17242116
Ball Radius, r (m)	0.001755	0.00000795505

The following formula has been derived to calculate the error for Viscosity:

$$\Delta\eta = \sqrt{\left(\frac{\partial\eta}{\partial r}\Delta r\right)^2 + \left(\frac{\partial\eta}{\partial p_s}\Delta p_s\right)^2 + \left(\frac{\partial\eta}{\partial p_l}\Delta p_l\right)^2 + \left(\frac{\partial\eta}{\partial v}\Delta v\right)^2}$$

$$= \sqrt{((r*8829/25/(9*v)^2*(p_s-p_l)*v)\cdot\Delta r)^2 + ((r^2*8829/50/(9*v)^2*v)\cdot\Delta p_s)^2 + ((r^2*-8829/50/(9*v)^2*v)\cdot\Delta p_l)^2 + ((r*-8829/50/(9*v)^2*(p_s-p_l))\cdot\Delta r)^2}$$

=0.0659256657 Pascal-Seconds

Using the above formula with the obtained values, we have calculated the Viscosity of the Fluid to be **2.71 ± 0.0659 Pascal-Seconds (3sf)**

There are several possible reasons why the calculated viscosity of **2.71 ± 0.0659 Pascal-Seconds (3sf)** is higher than the expected value of **1.2 Pascal-Seconds** [1] for glycerine. Temperature fluctuations may have influenced the results, as glycerine is highly sensitive to heat, and impurities, such as water, could have increased its viscosity. Errors in measuring terminal velocity, ball radius, and densities likely contributed to the discrepancy. Additionally, inaccuracies caused by human error, such as timing mistakes, and issues with equipment calibration played a role. These factors collectively account for the difference and highlight the importance of improved precision and stricter experimental controls.

References

[1] Takamura, K., Fischer, H. and Morrow, N.R. (2012). Physical properties of aqueous glycerol solutions. *Journal of Petroleum Science and Engineering*, [online] 98-99, pp.50–60. doi:https://doi.org/10.1016/j.petrol.2012.09.003.